

MILE-HIGH EXPECTATIONS: WHAT REALLY SHAPES THE AIRLINE EXPERIENCE



Pilots:

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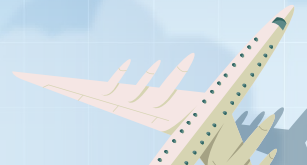
INTRODUCTION & PROBLEM STATEMENT

Overview of the Problem: Airline passengers frequently voice complaints about delays, discomfort, and poor service. Airlines often emphasize values like hospitality, innovation, and safety in their branding. Yet passenger reviews frequently focus on discomfort, delays, and poor service. This highlights a gap between airline priorities and customer experience.

Key question: What themes consistently emerge in airline reviews, and how do they contrast with how airlines present their values and priorities?

Why it matters: Understanding whether negative sentiment stems from operational issues (e.g. customer service, scheduling) or technical ones (e.g. cabin layout, aircraft age) can help airlines and manufacturers target improvements more effectively.

Stakeholders: Airlines, aircraft manufacturers, and travelers



DATA SOURCES, PRE-CLEANING, INTERESTING DETAILS

Data:

- Skytrax Airline Reviews
 - Scraped 10 pages for 20 Worldwide Airlines
 - Text Data, Ratings, Flight Types, etc.
- Airline Websites
 - Scraped for Core Values & Mission Statements

Pre-Cleaning:

- Filtered for English-language reviews only
- Removed duplicate reviews and standardized airline names (e.g., “Delta Airlines” vs “Delta Air Lines”)

Interesting Notes:

- Many airlines use similar phrasing on core values
 - Safety, Customer Satisfaction, Innovation, etc.
- Customer ratings were much more populated at high and low ratings (i.e. only having a really good/really bad experience)

9/10

"what I call customer service"

Bradley Martin (United States) 14th April 2025

Not Verified | I was late for my United flight from Harrisburg to Seattle due to traffic caused by an accident. The phone bot was clunky but once I got through to an agent they were able to reschedule my reservation even though it was a basic economy ticket. That's what I call customer service!

Type Of Traveller	Solo Leisure
Seat Type	Economy Class
Route	Harrisburg to Seattle
Date Flown	April 2025
Seat Comfort	★★★★☆
Cabin Staff Service	★★★★☆
Food & Beverages	★★★★☆
Inflight Entertainment	★★★★★
Ground Service	★★★★★
Wifi & Connectivity	★★★★☆
Value For Money	★★★★★
Recommended	✓

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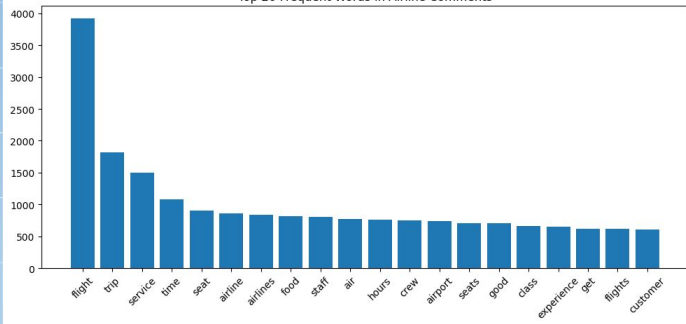
"one of the worst experiences"

Richard Hooks (United States) 12th April 2025

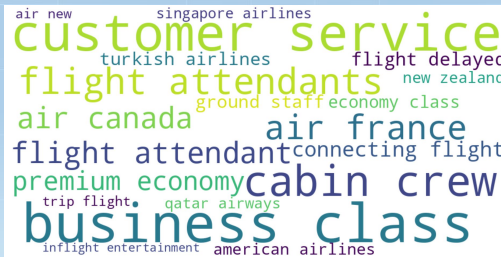
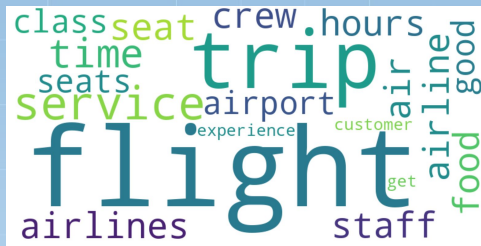
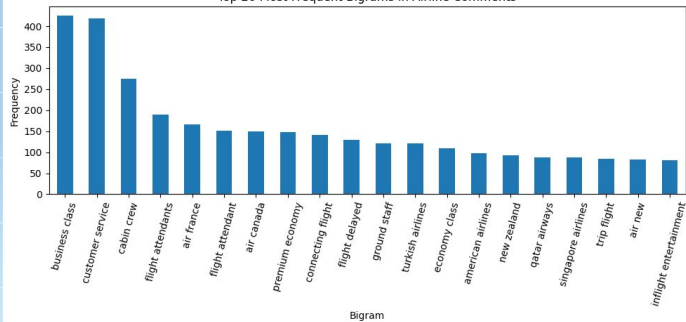
✓ *Trip Verified* | Absolutely one of the worst experiences on an airline. I flew out of Cebu to Seoul. I made the connection and United took off close to the time scheduled. The pilot announced about 1 1/2 hours into the flight we had to return to Seoul because of a problem with the distribution of the weight on the plane. They placed us on the Skybridge, no food or drinks, nowhere to sit, and we finally took off again later. The attendants were not very friendly and unprofessional. We finally arrived in San Francisco and I missed my connecting flight. I had to wait another 13 hours for another flight. No meal vouchers or hotel vouchers while waiting a long time for another flight. I finally landed in Raleigh Durham and then United had lost my luggage. I had to file a claim with United and the agent did not even know they had a flight coming in that morning. I had to spend a lot of time calling and trying to find out about my luggage. The customer service representative laughed about it on the phone. I actually inquired about my luggage when arriving in San Francisco and also about the missed connection and the customer service representative laughed about it. It took another week to finally get my luggage back from United Airlines.

MOST FREQUENT WORDS, TF-IDF, AND BIGRAM

Top 20 Frequent Words in Airline Comments



Top 20 Most Frequent Bigrams in Airline Comments



INSIGHTS

Customers frequently talk about **flights, service, and seats**. There are key experience touchpoints (food, staff, crew).

1. Service, food, and seating are considered **high-impact experience drivers**.
2. Several insights regarding **flight delays** and **cabin quality** are signal areas for improvement.

Customers highlight **customer service touchpoints** (customer service, cabin crew, flight attendants). Customers serve importance to **seating tiers** (business class, premium economy, economy class).

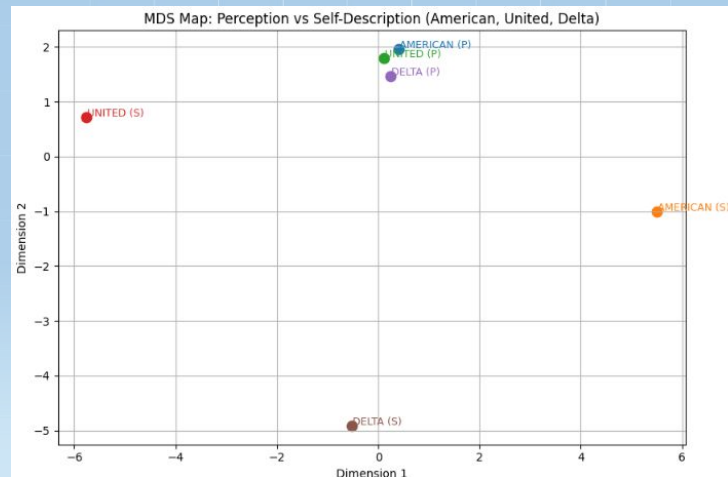
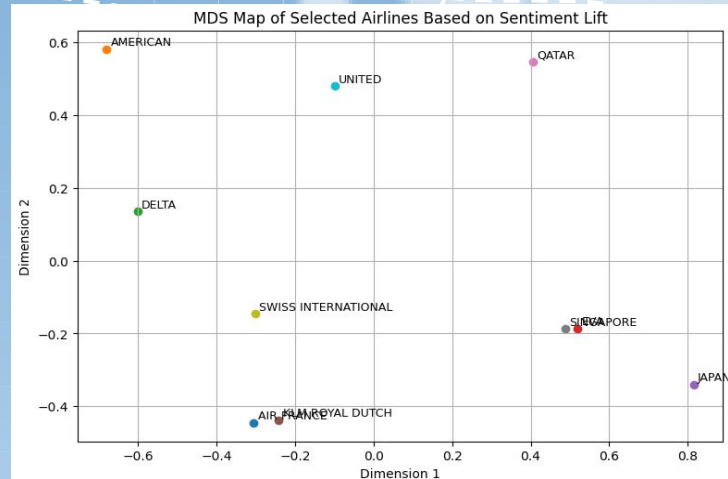
LIFT ANALYSIS & MDS MAPS

Sentiment Lift Matrix (Selected Airlines):

airline	SINGAPORE	EVA	JAPAN	AIR FRANCE	KLM ROYAL DUTCH	SWISS INTERNATIONAL	AMERICAN	UNITED	DELTA	QATAR
neg	0.643	0.617	0.386	1.311	1.260	1.260	1.465	1.003	1.465	0.591
neutral	0.746	0.746	0.597	0.597	0.597	0.896	1.642	1.493	1.194	1.493
pos	1.287	1.305	1.489	0.827	0.864	0.827	0.588	0.938	0.643	1.232

INSIGHTS

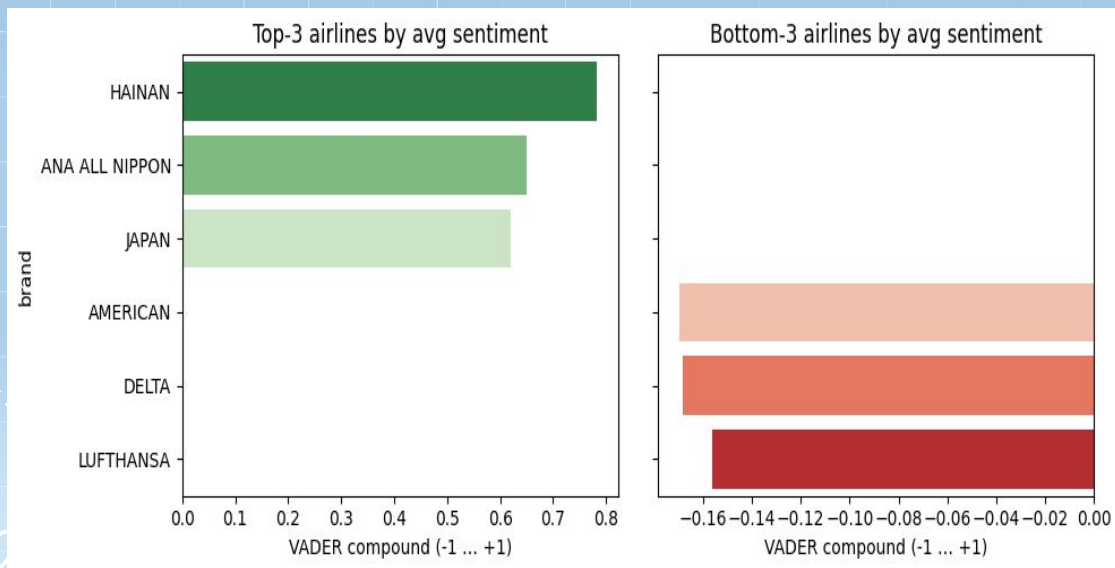
1. AIR FRANCE, KLM ROYAL DUTCH, and SWISS INTERNATIONAL form one cluster; JAPAN, EVA, and SINGAPORE form another.
2. SWISS INTERNATIONAL sits near the center, suggesting an average sentiment profile.
3. Airlines on the right have higher positive lift; those at the bottom have lower neutral lift.
4. The three domestic airlines' self-descriptions differ from public perception.



SENTIMENT ANALYSIS & TOPIC MODELING

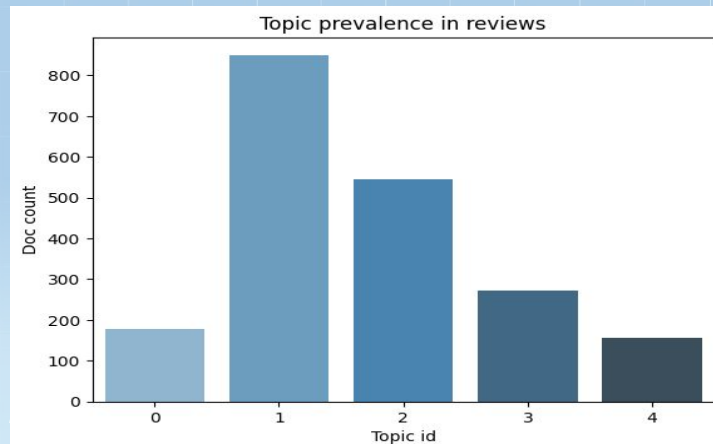
Airline sentiment scoreboard

- Winners: Hainan (+0.78), ANA All Nippon (+0.65), Japan Airlines (+0.61) – clear positive brand advocates; current service formula is working.
- Under-pressure: American (-0.17), Delta (-0.15), Lufthansa (-0.16) – net-negative tone; 0.95-pt gap to the leader highlights a reputational hit worth addressing.



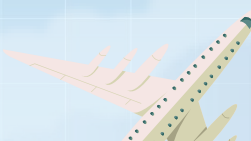
What customers actually talk about

- Topic 1 dominates (45 % of all comments) – keywords around “flight, verified, trip” ⇒ travellers obsess over end-to-end flight execution and status accuracy.
- Next two themes (Topic 2 & 3, 34 %) – “service/crew” and “time/delay” reveal staff interaction and punctuality as the next biggest levers.
- Long-tail (Topic 0 & 4, 21 %) – seat-fee frustration and route-specific gripes like Toronto are smaller but rising early-warning signals.





LIGHTBULB MOMENTS

1. There's a serious **gap between brand narrative and brand experience** and customers are noticing.
 - Brands like American and Delta have the most customer-centric core values, but show some of the **most negative sentiment lifts**.
 2. Region predicts **customer experience**, indicating that regional norms/service cultures shape customer perception.
 - **Clear clustering by geography**, even with region input not being a model feature.
 3. Customers care less about perks and more about performance.
 - 45% of review content is about **verification, timing, and logistics - not food and entertainment**.
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KEY INSIGHTS & ACTIONABLE RECOMMENDATIONS

Insight

- 1. Brand-Experience Mismatch:** U.S. airlines (American, Delta, United) and Lufthansa show high negative sentiment lift, despite customer-first brand messaging. A 0.95-pt sentiment gap to leaders like Hainan (+0.78) signals reputational risk.
- 2. Regional Clustering Patterns:** Airlines cluster by geographic region in sentiment space; Asian carriers (ANA, JAL, Hainan) show strong positive lift.
- 3. Key Experience Drivers:** Service, food, and seating consistently dominate review content and bigrams—clear high-impact levers.
- 4. Generic Messaging:** Airline core values often rely on generic language (“innovation,” “satisfaction”), missing what customers actually value.
- 5. Flight Execution Is Central:** Topic modeling shows 45% of reviews are focused on end-to-end flight status, accuracy, and disruptions—this is the emotional center of the flying experience.

Recommendation

- Recalibrate Messaging or Improve Delivery:** Align brand with actual experience—prioritize frontline training, delay recovery, and service consistency.
- Benchmark by Region:** Study top regional performers to identify replicable practices—on-time ops, staff tone, cabin quality.
- Prioritize High-Impact Touchpoints:** Invest in seat comfort, meal quality (esp. in economy), and consistent crew service.
- Differentiate Through Real Experience:** Revise brand language to reflect unique praise (e.g., “attentive staff,” “reliable flights”) surfaced in reviews.
- Reassure on Operational Reliability:** Highlight real-time flight updates, punctuality, and smooth trip execution in marketing and strategy.

FUTURE WORK

1. What We Didn't Get to Do

- **Aircraft-Level Ratings:** Most data is airline-level. Pairing seat maps, aircraft age, or cabin configurations would improve attribution accuracy.
- **Integrate Social Media Feedback:** Access Reddit or Twitter APIs to expand beyond Skytrax reviews to include more customer feedback.

2. Taking the Work Further

- **Collaborate with Airlines:** Partner with carriers to validate findings and inform service design or fleet decisions.
- **Interactive Visuals for Passengers:** Build a public-facing dashboard for flyers to explore sentiment by airline, aircraft type, or route before booking.

Q&A

